Daromad

Income evidence for borrowers who have no payslip

Mode: Partner product — MKBANK, micro & family-business credit department

Team TYS Munisbek (technical lead) · Jahongir (research & data) · Umida (compliance & business case)

Live prototype https://<your-project>.vercel.app **Repository** https://github.com/hmunisbekk-droid/TYS-fintech

FinTech / AI in Finance · partners MKBANK and the Central Bank of Uzbekistan · final defence, October 2026

The regulation names the evidence — the self-employed have exactly one line

CBU regulation on the debt-burden ratio, cl.9 — income a lender may rely on

- Employer-confirmed salary — does not exist: income untaxed, no returns, no receipts required
- Pension — does not exist: income untaxed, no returns, no receipts required
- Funded-pension contributions — does not exist: income untaxed, no returns, no receipts required
- ✓ **Income credited to the borrower's bank accounts — the only one a self-employed borrower has**
- Taxes paid — does not exist: income untaxed, no returns, no receipts required
- Interest, dividends, rental income — does not exist: income untaxed, no returns, no receipts required

50% cap

ПДН limit for all loans since 1 Jan 2025 (cl.10); 12-month arithmetic mean of income (cl.7)

~140,000

micro-projects MKBANK targets in 2026 (47,975 in 2025); 1,062 mahalla agents

60–90 min

an officer's manual reading of 12 months of statements per application (our estimate; pilot measures it)

0 lines

of official income records for ~790k registered self-employed (untaxed, unfiled)

Sources: lex.uz/docs/4654149 and 6809992 (CBU regulation and 2024 amendments) · spot.uz 27 Mar 2026 (MKBANK, bank's own figures) · yuz.uz (self-employed registrations, 2021) · buxgalter.uz

What the officer gets in under five seconds



12-month regulatory income

cl.7 mean, cl.9 definition; own-card transfers, loan proceeds, refunds and cash deposits excluded

ПДН with citations

36-month spreading, guarantor rule, 50-BRV and family-programme exemptions, 50% cap, maximum affordable amount

Integrity report

balance chain, duplicates, coverage, gaps, pre-application spikes, circular transfers, holder match

Verified memo

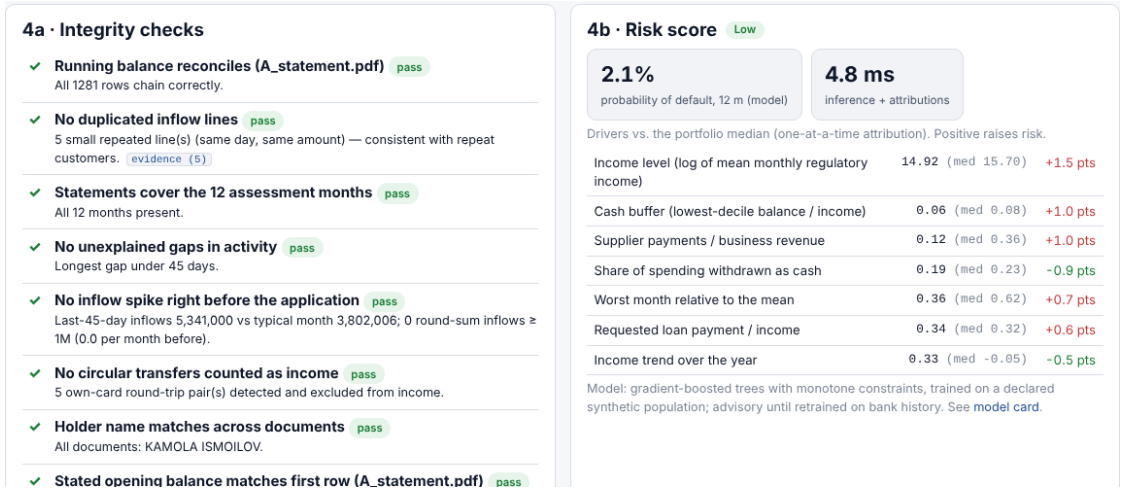
every number traceable to a document line; the verifier refuses unsourced numbers; hash-chained audit log

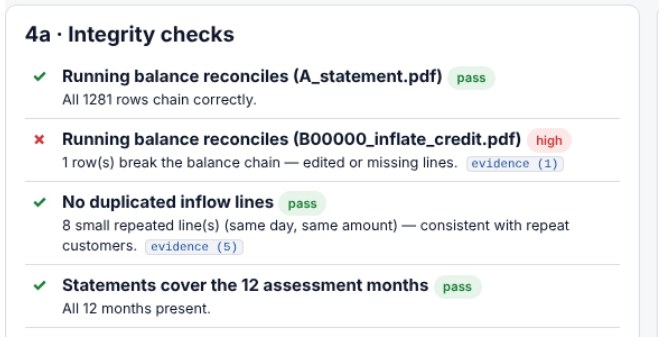
Live demo — on our samples and on anything you upload

Open on your laptop or phone

https://<your-project>.vercel.app

1. Load sample A — statement PDF, Soliq certificate, receipt photo parse in the tab (no server, no API)
2. Income series, ПДН with clause citations, maximum affordable amount; drag the slider to cross the 50% cap
3. Integrity checks and risk score with drivers; click any evidence link to see the statement line
4. Memo: verifier badge shows how many numbers were checked and that none is unsourced
5. Record the decision; export the case with its hash-chained audit log
6. Then: Load tampered statement — the balance chain breaks at the inflated inflow; upload your own documents





Scan for the repository; the live site link is on the title slide and in the README

Seven components, one optional language model, zero API keys

1 · Intake

pdf.js with coordinates; XLSX/CSV; tesseract.js OCR uzb+rus+eng, self-hosted; field confidence, tap-to-confirm

2 · Categoriser

multilingual rules + counterparty patterns (recurring payer → family; same mask both ways → own card)

3 · Cash-flow

12-month series; 18 features identical in TypeScript and Python (parity test)

4 · Rule engine

ПДН clause by clause: 36/180-month spreading, guarantor rule, exemptions, 50% cap, back-solved amount

5 · Risk model

XGBoost, monotone constraints, JSON trees evaluated in the browser; attributions vs portfolio median

6 · Integrity

balance chain, duplicates, coverage, gaps, round-sum spikes, circular transfers, holder match

7 · Memo + audit

template from a fact table; verifier rejects unsourced numbers; SHA-256 hash-chained log

Perimeter

Prototype: static site, every byte of a document stays in the browser. Production: the same bundle on the bank's server, SSO, audit log in the bank DB, core-banking pull for own customers. LLM optional, narrative only, behind the same verifier.

Speed: statement 1.2–1.8 s · receipt OCR 1.6 s · everything else < 50 ms · marginal cost per application: zero

Human in the loop: the engine recommends and computes the maximum affordable amount; the officer decides; an override needs a reason, written into the chain.

Measured through the real app, not typed into a slide

60/60

statement PDFs parsed with exact row count and reconciled balance chain (2 layouts, RU/UZ)

40/40

tampered statements caught: inflated inflow, deleted debit, duplicated row, window dressing

95%

receipt OCR — total amount exact on 60 camera-augmented photos; INN 100%, date 97%

99.96%

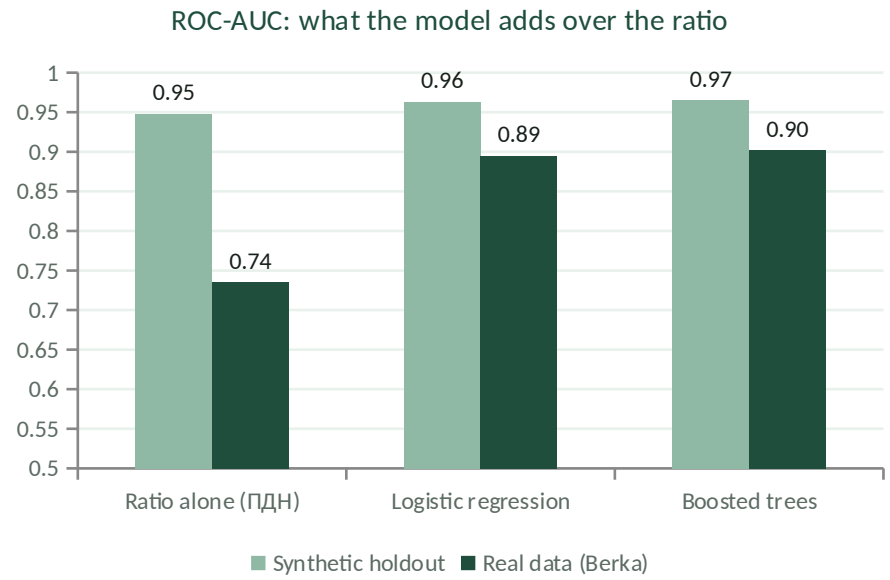
categoriser vs ground truth on 10,481 lines — synthetic vocabulary; expect lower on real data

0.965

risk model ROC-AUC, synthetic holdout n=500; logistic 0.963, ratio alone 0.948

0.902

ROC-AUC on real anonymised bank data (Berka, 443 loans, 5-fold) vs 0.735 for the ratio alone



Honest reading: on synthetic data the ratio already carries most of the signal by construction; on real statements the cash-flow features add +0.17 AUC over the ratio. Memo faithfulness: 0 unsourced numbers on every sample case (enforced at run time).

Statement parsing and tamper detection (through the real app)

100.0% row count exact, 60 PDFs (2 layouts, RU/UZ)	100.0% balance chain reconciles
1.6 s per 12-month statement	100.0% tamper recall, 40 tampered PDFs

tamper type	n	detected
inflate_credit	10	10
delete_debit	10	10
duplicate_row	10	10
window_dressing	10	10

Synthetic statements rendered by headless Chromium (two layouts, RU and UZ labels), parsed by the app's pdf.js pipeline; tampered variants from synth/tamper.py.

Receipt OCR (tesseract.js, in-browser)

95.0% total amount exact, n=60	100.0% INN exact	96.7% date exact
1.6 s per photo · mean confidence 81.9%		

Synthetic receipts rendered to PNG, then camera-augmented (perspective, rotation, lighting, blur, JPEG). Measured through the real app with tesseract.js uzb+rus+eng.

All numbers are read from JSON produced by ml/train.py, ml/berka_validate.py, npm test and tests/*_eval.py (see the Model card page).

Caveat on the slide, not in the footnote: the risk score is advisory until retrained on the bank's own history inside the perimeter — that is the pilot's first deliverable.

No partner data, so we built a declared generator

1

Archetypes

10 trades (tailor, taxi, dehkan farmer, grocer, reseller, tutor…) with declared seasonality, customer counts, channel mix, expense structure

2

Simulation

2,000 borrowers · 5,689,589 transactions · existing loans with annuity schedules · shocks · default label from a latent affordability score (10.5% rate)

3

Rendering

60 statement sets as PDF (text layer) + XLSX + CSV in RU and UZ layouts; 180 fiscal receipts; Soliq-style certificates with QR

4

Camera model

perspective, rotation, lighting gradient, blur, sensor noise, JPEG — then measured through the same OCR the officer uses

5

Tampering

40 statements: inflated inflow, deleted debit, duplicated row, window dressing — labelled for the integrity evaluation



HUMO CARD STATEMENT
Выписка по карте

Владелец: KAMOLA ISMOILOV
Карта: 5614 **** **** 4369
Период: 2025-09-01 — 2026-08-31
Валюта: UZS

Входящий остаток: **668 582.52**

Дата	Время	Описание операции	Расход	Приход	Остаток
2025-09-01	07:33	P2P HUMO 8600 **** 9613		50 000.00	718 582.52
2025-09-01	08:33	IJARA TO'LOVI P2P 8600 **** 2210	674 000.00		44 582.52
2025-09-01	08:54	NAQD PUL KIRITISH BANKOMAT		350 000.00	394 582.52
2025-09-01	09:03	HUDUDGAZTAMINOT KOMMUNAL	131 000.00		263 582.52
2025-09-01	14:57	HUDUDIY ELEKTR TARMOQLARI KOMMUNAL	90 000.00		173 582.52
2025-09-01	16:24	UCELL ABONENT	62 000.00		111 582.52
2025-09-02	14:09	BARAKA MARKET TOSHKENT	103 000.00		8 582.52
2025-09-03	16:36	POS KAMOLA SARTAROSHLIK TERMINAL		52 000.00	60 582.52
2025-09-04	07:29	P2P HUMO 9860 **** 9719		17 000.00	77 582.52
2025-09-04	12:45	POS KAMOLA SARTAROSHLIK TERMINAL		40 000.00	117 582.52
2025-09-04	13:46	POS KAMOLA SARTAROSHLIK TERMINAL		22 000.00	139 582.52
2025-09-04	16:24	P2P HUMO 8600 **** 6177		38 000.00	177 582.52
2025-09-04	16:48	P2P HUMO 9860 **** 4617		39 000.00	216 582.52
2025-09-04	17:40	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	243 582.52
2025-09-04	22:21	ANGLESEY TOSHKENT	63 000.00		180 582.52
2025-09-05	20:06	POS KAMOLA SARTAROSHLIK TERMINAL		32 000.00	212 582.52
2025-09-05	20:29	P2P HUMO 8600 **** 9240		31 000.00	243 582.52
2025-09-05	22:57	OPTOM SAVDO MCHJ TO'LOV	104 000.00		139 582.52
2025-09-06	14:53	POS KAMOLA SARTAROSHLIK TERMINAL		64 000.00	203 582.52
2025-09-06	17:24	HAVAS TOSHKENT	131 000.00		72 582.52
2025-09-07	11:40	P2P HUMO 9860 **** 3683		66 000.00	138 582.52
2025-09-07	13:08	POS KAMOLA SARTAROSHLIK TERMINAL		31 000.00	169 582.52
2025-09-07	20:29	OPTOM SAVDO MCHJ TO'LOV	88 000.00		81 582.52
2025-09-08	07:44	POS KAMOLA SARTAROSHLIK TERMINAL		43 000.00	124 582.52
2025-09-08	15:53	POS KAMOLA SARTAROSHLIK TERMINAL		39 000.00	163 582.52
2025-09-08	16:34	OPTOM SAVDO MCHJ TO'LOV	106 000.00		57 582.52
2025-09-08	18:41	KORZINKA TOSHKENT	42 000.00		15 582.52
2025-09-09	08:34	POS KAMOLA SARTAROSHLIK TERMINAL		50 000.00	65 582.52
2025-09-09	09:50	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	92 582.52
2025-09-09	21:57	HAVAS TOSHKENT	50 000.00		42 582.52
2025-09-10	12:30	POS KAMOLA SARTAROSHLIK TERMINAL		30 000.00	72 582.52
2025-09-10	18:43	POS KAMOLA SARTAROSHLIK TERMINAL		46 000.00	118 582.52
2025-09-10	22:08	P2P HUMO 8600 **** 3739		34 000.00	152 582.52
2025-09-11	10:18	P2P HUMO 9860 **** 5631		33 000.00	185 582.52
2025-09-11	17:02	POS KAMOLA SARTAROSHLIK TERMINAL		37 000.00	222 582.52
2025-09-11	18:42	P2P HUMO 8600 **** 7135		44 000.00	266 582.52
2025-09-11	21:26	P2P HUMO 9860 **** 6338		30 000.00	296 582.52
2025-09-11	22:12	P2P HUMO 9860 **** 6674		34 000.00	330 582.52
2025-09-12	15:07	POS KAMOLA SARTAROSHLIK TERMINAL		27 000.00	357 582.52
2025-09-12	18:20	P2P HUMO 9860 **** 2803		24 000.00	381 582.52



KAMOLA SARTAROSHLIK
Xorazm, Mustaqillik ko'chasi 9
STIR/INN: 342056057
FISKAL CHEK № 58332
01.03.2026 13:36
Kamola
Sana
Kassir
BOSH YUVISH 45 000
3 x 15 000
SOCH OLISH 40 000
1 x 40 000
JAMI 85 000
To'landi (UZCARD) 85 000
QQS 12% 9 107
FM: UZ118914450618 FP: 135878439006
ofd.soliq.uz
Xaridingiz uchun rahmat!

Synthetic statement (Humo-style layout, Russian labels) and a camera-augmented synthetic fiscal receipt. No real person is represented; names come from a short public list, card numbers are random and masked.

```
Regenerate:  python -m synth.cli --n 2000 --render 60 --
receipts 180 --tamper 40 --seed 42
```

Also: methodology check on real anonymised bank data (Berka, PKDD'99) — not redistributed, downloaded by script.

Pilot: three months, five branches, fifty mahalla agents

Cost to build and run the pilot (assumptions, UZS)

Line	Assumption	3-month cost
Team: 3 engineers/analysts (TYS)	3 × 3 months × 15M	135M
Bank credit analyst, part-time	0.5 FTE × 3 months × 15M	22.5M
Infrastructure	1 VM in the bank DC, no GPU	≈ 3M
Integration (core banking, bureau)	bank IT, 20 person-days	≈ 30M
Data, licences, API spend	none — open-source, no external API	0
Total pilot (≈ $15k)		**≈ 190M**

Running cost per application: zero marginal (static bundle; no API, no GPU). Optional narrative LLM on one on-prem GPU: well under 1 US cent per application.

Timeline: month 1 shadow mode (engine runs beside the manual process; we measure minutes and agreement) · months 2–3 assisted mode (officer works from the memo) · success metrics: minutes per application, agreement with committee, override rate, ПДН audit findings, retrained model AUC on bank history.

Value — the officer-hours formula (every term is a slider)

90 → 25 min

document handling per application, before → with Daromad (assumption; pilot measures it)

140,000

micro-project applications per year — MKBANK's own 2026 target

≈ 150,000 h

officer-hours saved per year = 140,000 × 65 min

≈ 75 FTE-years

at 2,000 working hours per FTE ≈ 1.3 bn UZS/yr at 15M/month loaded cost

Payback: pilot cost ≈ 190M against ≈ 1.3 bn/yr of officer time at full roll-out → under 3 months after roll-out; even at one-tenth of the time saving, under a year. Not counted: faster time-to-decision for the borrower, fewer ПДН compliance findings, and better risk selection — which we will only claim after the pilot measures it.

Who pays: MKBANK (credit department / digital budget). Who owns: micro & family-business credit department; IT deploys; mahalla agents and loan officers use it.

Hard to copy in a weekend — and what comes next

The replication test — what the same LLM tools do not give you

- Statement parsers that learn columns from the data rows (two Uzbek layouts today, any text-layer PDF, XLSX, CSV) and reconcile the balance chain
- A categoriser lexicon in Uzbek Latin, Cyrillic and Russian with counterparty-pattern rules — and the pilot plan to grow it on bank data
- A rule engine mapped clause by clause to the live ПДН regulation, with its exemptions and the back-solved maximum amount
- A memo verifier that makes unsourced numbers impossible — zero hallucination by construction, no LLM needed
- A declared, calibrated synthetic population with rendered documents, a camera model and labelled tampering — a test bed the bank can keep using
- Evaluation through the real app, reproducible with one command each

Next steps

- Week 1–2: scanned-statement OCR page by page; lexicon growth on anonymised bank descriptions; Uzbek/Russian memo templates
- Month 1: shadow mode in 5 branches; core-banking pull for MKBANK's own account holders; credit-bureau REST for obligations
- Month 2–3: assisted mode; retrain the risk model on bank history inside the perimeter; set risk bands with the bank's risk appetite; fairness review by region
- After the pilot: Central Bank conversation — the same engine computes ПДН uniformly for any lender, a supervisory check at zero marginal cost
- Product scope today: Ommabop 2, Express, Business Online 2 (ПДН applies) and family-programme loans (exempt from the cap, capacity still assessed)

Daromad

Income evidence for borrowers who have no payslip — regulation-native, perimeter-native, verified.

Live: https://<your-project>.vercel.app
Repo: https://github.com/hmunisbekk-droid/TYS-fintech
Team TYS: Munisbek · Jahongir · Umida

Appendix A · ПДН rule engine, clause by clause

Clause	Rule as implemented	Where it shows
cl.1	ПДН = average monthly payments on loans/microloans ÷ average monthly income (borrower + co-borrower)	formula panel
cl.2	Includes the application under review and all loan agreements with all lenders; co-borrower loans counted; guarantor loans only if > 30 days overdue; excluded if amount + outstanding principal ≤ 50 BRV	obligations table, exemption badge
cl.4[1]	Terms over 36 months: total payments spread over 36 (annuity); mortgages over 180 months spread over 180	per-line note
cl.6	Calculation date within 10 calendar days of the loan date	case date
cl.7	Average monthly income = arithmetic mean of the last 12 months; co-borrower income added	income base
cl.9	Evidence: employer salary data, pension, funded-pension contributions, income credited to bank accounts, taxes paid, interest/dividends/rent; other data allowed	memo section 2
cl.10	Ratio ≤ 50% from 1 January 2025 (60% between 1 July 2024 and 1 January 2025)	cap, pass/fail
cl.11	Family-entrepreneurship programme loans exempt from the ratio requirement	exemption badge

BRV (базовая расчётная величина) is a configuration value; update from lex.uz. Verified 10 Oct 2026.

Appendix B · Failure cases, perimeter, human in the loop

Known failure cases (also on the Model card page)

- Scanned statements without a text layer: not OCR'd page by page yet — ask for the XLSX/CSV export
- One signed amount column and no balance column: signs taken from keywords only
- Merchant strings outside the lexicon (mixed scripts, transliteration) → default category, visible in the table
- Dark or heavily skewed receipt photos: OCR confidence < 60% → fields flagged for confirmation, never guessed
- Cash-only businesses: nothing to reconstruct — the memo says so; the officer attests
- Synthetic-to-real gap of the risk score until retrained inside the bank; risk bands are placeholders

Perimeter, audit, human in the loop

- Prototype: static site; documents are parsed in the browser and never transmitted; nothing is stored server-side
- Production: same bundle on the bank's web server (Docker, no outbound network), SSO identity, audit log and cases in the bank DB
- Own customers' statements pulled from core banking; uploads only for other banks' statements and wallets
- Every analysis stamped with rule-engine, model and categoriser versions; hash-chained SHA-256 log; export with the case
- The engine recommends; the officer decides; overrides require a reason; committee above branch limits
- Personal data processed inside the bank with recorded consent; retention per bank policy; no gender or activity-type features in the model

Appendix C · Data sources and licences

Source	Used for	Licence / terms
Daromad synthetic data set (synth/, seed 42)	training, evaluation, demo samples	MIT, this repo; generator and parameters declared; no real person
Berka / PKDD'99 Financial data set (mirror github.com/jlacko/berka-dataset)	offline methodology check	academic challenge data, no licence stated; evaluation only, not redistributed
Tesseract tessdata_fast (uzb, uzb_cyrl, rus, eng); tesseract.js	receipt OCR	Apache 2.0
pdf.js; SheetJS; Next.js; React	parsing, application	Apache 2.0 / MIT
XGBoost; scikit-learn; NumPy; pandas; OpenCV; Playwright; Jinja2; qrcode	training, rendering, evaluation	Apache 2.0 / BSD-3 / MIT
CBU regulation texts (lex.uz 4654149, 6809992)	rule engine, citations	public legal information
stat.uz wages; CBU statistics; MKBANK public pages (sponsored article, bank's own figures)	calibration anchors, business case	public information, cited

Rules compliance: no hard-coded answers (every figure computed from uploaded documents at run time); no real personal or confidential data; all code written during the final by team TYS with LLM coding assistance (declared); deployment from the submitted repository; CI on every push.